Commutative Property - Definition | Commutative Law and Examples

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The **Commutative property** is a basic math rule that helps make calculations easier. It states that the result of an operation between two numbers remains the same irrespective of the position of the numbers. For example, 2 + 3 is the same as 3 + 2, and 4 × 5 is the same as 5 × 4.

It is an important property of mathematics. Which is satisfied by the addition(+) and multiplication(×) operations.

In this article, we will explore the commutative property, its definition, and examples, in detail.

What is Commutative Property?

Commutative Property is the property according to which the sequence of two operands does not change the result. In commutative property the switching of the operands does not affect the result of the given expression.

The commutative property is the property in which the order of the two operands does not affect the result of the expression. The commutative word is derived from the word commute which means switching. The arithmetic operator addition and multiplication satisfies the commutative property.

Commutative Property Definition

Examples,

- 4 + 7 = 7 + 4 = 11
- 5 + 4 = 4 + 5 = 9
- 4 × 7 = 7 × 4 = 28
- 5 × 4 = 4 × 5 = 20

Commutative Property Formula

Commutative property is satisfied by the addition and multiplication arithmetic operators only. For two numbers P and Q, the commutative property formula is given by:

- ***P + Q = Q + P***
- ***P × Q = Q × P***

Commutative Property of Addition

According to the Commutative property for addition the order of two operands related with addition operator does not affects the result of addition.

Commutative Property for Addition Formula

For addition of two numbers X and Y, the commutative property for addition formula is given below:

X + Y = Y + X

Commutative Property of Addition Examples

Following are some examples for the commutative property for addition:

- 4 + 9 = 9 + 4
- (-12) + 5 = 5 + (-12)
- 8/9 + 1/3 = 1/3 + 8 / 9

Commutative Property of Multiplication

The commutative property for multiplication states that the order of two operands related with multiplication operator has no affect in the result of multiplication.

Commutative Property for Multiplication Formula

For multiplication of two numbers X and Y, the commutative property for addition formula is given below:

X × Y = Y × X

Commutative Property of Multiplication Examples

Following are some examples for the commutative property for addition:

- 4 × 9 = 9 × 4
- (-12) × 5 = 5 × (-12)
- (8 / 9) × (1 /3) = (1 / 3) × (8 / 9)

Non-Commutative Operations - Division And Subtraction

Non-Commutative operations refers to those operations that do not follow the commutative property and changing the order of the numbers in the operations changes the result of the operation. The arithmetic operators subtraction and division does not satisfy the commutative property as changing the order of the operands changes the result of the expression, and this can be explained by the given examples.

For two operands R and S,

- R - S ≠ S - R
- R / S ≠ S / R

Example 1: Suppose we take two numbers 12 and 3 then dividing 12 by 3 and dividing 3 by 12 gives the separate results, i.e.

- 12/3 = 4
- 3/12 = 1/4

Example 2: Let's take two numbers 15 and 35 then subtracting 15 from 35 and subtracting 35 from 15 gives the separate results, i.e.

- 35 - 15 = 20
- 15 - 35 = -20

Commutative Property vs Associative Property

The differences between commutative property and associative property is explained in the table below,

Commutative Property	**Associative Property**
Commutative property states that the order of the number in some operations(such as multiplication and addition) dose not effects the results of the operation.	Associative property states that grouping various number in different operations (such as multiplication and addition) does not changes the result of the operation.
Commutative Formula: • A + B = B + A • A × B = B × A	**Associative Formula:** • A + (B + C) = (A + B) + C = (A + C) + B • A × (B × C) = (A × B) × C = (A × C) × B
Example: 4 + 5 = 5 + 4 = 9	**Example:** (3 + 4) + 5 = 3 + (4 + 5) = 12

Read More,

- *Properties of Rational Numbers*
- *Properties of Real Numbers*
- *Properties of Integers*

Solved Examples on Commutative Property

Example 1: Which of the following satisfies the commutative property:

(i) 4 × 5

(ii) 7 + 8

(iii) 9 / 2

(iv) 15 - 6

Solution:

- ***4 × 5***

4 × 5 = 20

5 × 4 = 20

Commutative Property Satisfies

- ***7 + 8***

7 + 8 = 15

8 + 7 = 15

Commutative Property Satisfies

- ***8 / 2***

8 / 2 = 4

2 / 8 = 1 / 4

4 ≠ 1/4

Commutative Property Does Not Satisfies

- ***15 - 6***

15 - 6 = 9

6 - 15 = -9

9 ≠ -9

Commutative Property Does Not Satisfies

Example 2: Prove that p + q = q + p if p = 9 and q = 7.

Solution:

p + q = 9 + 7 = 16

q + p = 7 + 9 = 16

p + q = q + p

Hence proved

Example 3: Prove that p × q = q × p if p = 10 and q = 3.

Solution:

p × q = 10 × 3 = 30

q × p = 3 × 10 = 30

p × q = q × p

Hence Proved

Practice Problems On Commutative Property

Problem 1: **Which of Following Satisfies Commutative Property**

42 × 15

12 + 7

23 / 3

25 - 8

Problem 2: Prove x + y = y + x if x = 10 and y = 17

Problem 3: Prove x × y = y × x if x = 2 and y = 23

Problem 4: Apply the commutative property of addition to rearrange and simplify the expression : 3a + 4b + 2a + 5b.

Conclusion

The commutative property is a helpful math rule that makes adding and multiplying numbers easier. It means you can change the order of the numbers, and the answer will stay the same! For example, 2 + 3 is the same as 3 + 2, and the same rule applies for multiplication 5 **×** 6 = 6 **×** 5. But remember, this doesn't work for subtraction or division. Knowing this property is super useful when solving math problems and makes math feel a lot less tricky

Commutative Property-FAQs

What is Commutative Property in Math?

Commutative property in Math is the property in which the switching of the two operands does not change the result of the expression.

What is Commutative Law for Multiplication?

Commutative property of multiplication states that the change in position of the two operands for multiplication does not change the result.

A × B = B × A

What is Commutative Law for Addition?

Commutative property of addition states that the change in position of the two operands for addition does not change the result.

A + B = B + A

Can Commutative Property be used for Subtraction and Division?

No, the commutative property cannot be used for subtraction and division as these operators does not satisfy the commutative property.

What do you Mean by Commutative Laws?

Commutative laws include the commutative property for addition and commutative property of multiplication.

- ***A + B = B + A***
- ***A × B = B × A***

What is Commutative Property for Addition?

Commutative Property for Addition states that, adding two numbers the order of the numbers in addition is not important, i.e. ***a + b = b + a.***